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RESEARCH ARTICLE

Stall Prediction in Quadcopters With SHAP-Based Explainability and a Novel Flight Dynamics Dataset

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ABSTRACT Uncrewed Aerial Vehicles (UAVs), particularly quadcopters, are increasingly deployed in high-stakes operations including defense, tactical surveillance, disaster response, and GPS-denied missions. However, aerodynamic stall—caused by conditions like vortex ring state or rotor stall—remains a critical failure mode that can lead to sudden lift loss and catastrophic crashes. This study presents a hybrid stall prediction framework that unifies physics-informed synthetic data generation, interpretable machine learning, and embedded rule-based logic for real-time, onboard risk assessment. A custom simulator was developed to generate 25,921 stall-prone flight states by modeling key parameters such as throttle, vertical speed, blade angle of attack (AoA), airspeed, disc loading, and thrust-to-weight ratio. Using this dataset, an XGBoost classifier was trained for binary classification with two output classes: stall and non-stall, and optimized via threshold tuning and SMOTE-based class balancing. Using this dataset, an XGBoost classifier achieved 0.97 precision, 0.98 recall, 0.98 F1-score, and 0.96 accuracy for stall prediction after integration with domain-specific rules. SHapley Additive exPlanations (SHAP) values were applied to provide transparent, instance-level justifications, revealing that throttle, VRS, and disc loading were consistently the top contributors to stall predictions. In parallel, domain-specific aerodynamic rules—such as critical AoA and high disc loading thresholds—were deployed to catch edge cases such as near-threshold predictions, ambiguous vortex ring state regions, or unseen aerodynamic configurations. This modular system is designed for compatibility with onboard processors like Raspberry Pi, enabling real-time deployment even in resource-constrained environments. It supports frugal innovation without compromising safety, making it suitable for stealth drones, ISR swarms, payload-carrying UAVs, and companion-computer-based quadcopters. By combining data-driven predictions with physics-aware rule triggers, the system reduces false negatives and false positives, improving reliability in mission-critical conditions. Ultimately, this work pushes drone autonomy toward proactive, explainable, and hardware-efficient safety mechanisms—where UAVs not only predict aerodynamic stall but also justify it in real time.

INDEX TERMS Quadcopter, stall prediction, Explainable AI (XAI), aerodynamic safety systems, SHAP, dataset.

I. INTRODUCTION

Quadcopters have emerged as indispensable tools across domains such as aerial surveying, infrastructure inspection, agriculture, and disaster response due to their vertical takeoff and landing capabilities, agility, and relatively low cost [1], [2]. However, their aerodynamic stability is highly sensitive to rapid changes in flight conditions—especially during aggressive maneuvers, steep descents, or when operating near maximum payload. In such regimes, one of the most critical

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and dangerous phenomena is aerodynamic stall: a sudden loss of lift caused by excessive blade angle of attack or disturbed airflow [3]. Unlike in fixed-wing aircraft, stall in quadcopters can occur asymmetrically or during low-speed hover, making it harder to detect and significantly more challenging to recover from in real-time scenarios.

Although aerodynamic stall prediction has been studied extensively in the context of fixed-wing platforms and wind turbines, existing approaches often depend on controlled wind tunnel conditions, steady-state assumptions, or computational fluid dynamics simulations that are not directly applicable to multirotor platforms. Furthermore, most machine learning models developed for flight anomaly detection act as closed boxes—lacking transparency and real-time interpretability [4], [5], which are crucial for mission-critical and safety-sensitive applications involving drones. To address these gaps, this study proposes a hybrid stall prediction framework specifically designed for quadcopters. The framework is built upon a supervised XGBoost classifier trained on a custom dataset collected using physics-informed synthetic data generation. This dataset captures time-series telemetry data including altitude, velocity, throttle, blade angle of attack, and IMU readings under both stable and pre-stall conditions. By incorporating flight logs from a variety of stall-inducing scenarios—such as aggressive descents, high payload stress, and sudden yaw changes—the dataset reflects diverse flight dynamics crucial for robust stall prediction.

Post-training, threshold tuning is applied to optimize predictive performance in the presence of class imbalance. To ensure model transparency, the system integrates SHAP (SHapley Additive exPlanations), providing instance-level interpretability and revealing how each input feature contributes to individual predictions. Further enhancing safety, a parallel rule-based layer—grounded in aerodynamic principles—monitors critical thresholds for Blade Angle of Attack, Throttle, and Vortex Ring State (VRS), issuing redundant warnings even when the ML model's confidence is low. This hybrid approach bridges the gap between closed-box AI and real-time explainable flight safety, delivering both predictive accuracy and operational clarity for onboard stall prevention in quadcopters.

II. THEORY

In aerial vehicles such as quadcopters, stall, as shown in Figure 1, refers to the loss of lift resulting from the airflow separation over the propeller blades or body due to an excessively high angle of attack (AoA). While traditionally associated with fixed-wing aircraft, stall in multirotors can occur during aggressive maneuvers, sudden wind gusts, or high-angle flight paths, where the airflow no longer remains laminar over the surfaces responsible for lift generation. The angle at which this phenomenon begins—termed the stall angle—is critical for ensuring safe and efficient flight, especially in autonomous or high-performance drones. In this study, we approach stall prediction as a classification problem by collecting a dataset comprising flight parameters such as attitude (pitch, roll, yaw), angular velocity, airspeed, and motor response under varying conditions.

A. FACTORS AFFECTING STALL

1) THROTTLE

Throttle determines the rotational speed of the propellers by controlling the power delivered to the motors. An increase in throttle increases rotor RPM and therefore lift, but also raises the blade's effective angle of attack. Excessively high throttle values, especially during aggressive ascent or descent, can lead to conditions like rotor stall or Vortex Ring State (VRS) [6]. Since lift L is proportional to the square of the rotor speed ω, i.e., $L \propto \omega^2$, throttle must be managed to prevent exceeding critical aerodynamic limits.

2) BLADE ANGLE OF ATTACK (AoA)

The angle of attack is the angle between the blade chord line and the relative airflow. Lift increases with AoA up to a critical limit (typically around $10^\circ - 15^\circ$), beyond which the airflow separates and the blade stalls [7]. This relationship is initially linear and given by $C_L \approx a \cdot \alpha$, where C_L is the lift coefficient and α is the AoA. Once the AoA exceeds the stall angle, the lift drops sharply. AoA can be indirectly affected by throttle, pitch angle, and flight velocity.

3) AIRSPEED

Airspeed influences the relative flow over the rotor blades. Low airspeed leads to an increased effective AoA for a given pitch [8], thereby increasing the risk of stall. In contrast, during high-speed forward flight, advancing blades may experience increased lift while retreating blades may stall, leading to asymmetric lift and instability. Therefore, airspeed plays a dual role in influencing both symmetric and asymmetric stall characteristics in quadcopters [9].

4) ALTITUDE

At higher altitudes, air density (ρ) decreases, which reduces lift, given by $L = \frac{1}{2}\rho V^2 S C_L$. To compensate for this, the quadcopter often increases throttle, which can inadvertently raise AoA and increase the risk of stall [10]. Thus, operating at high altitudes inherently brings the flight conditions closer to the stall regime, especially under loaded or aggressive flight scenarios.

5) BATTERY VOLTAGE

Battery voltage affects the available motor power. A drop in voltage leads to reduced rotor speed and may limit the quadcopter's ability to maintain lift. This becomes critical during rapid maneuvering or recovery from descent. Undervoltage conditions may force the system to operate at higher throttle percentages, inadvertently increasing AoA and leading to a stall condition.

FIGURE 1. Illustration of a quadcopter entering stall due to high descent rate and vortex ring state (VRS), progressing from stable flight to loss of lift.

6) DISC LOADING

Disc loading is defined as the total weight of the quadcopter divided by the total rotor disc area:

$$\text{Disc Loading} = \frac{W}{A}$$

Higher disc loading implies that each rotor must generate more lift per unit area, which usually requires higher RPM and AoA. This increases the aerodynamic stress on the blades, raising the likelihood of flow separation and stall [11]. Lightweight designs with lower disc loading are generally more stall-resistant.

7) THRUST-TO-WEIGHT RATIO (T/W)

The thrust-to-weight ratio indicates how much thrust the system can generate relative to its weight. A low T/W ratio implies that the system must operate closer to its maximum throttle during flight, increasing the risk of reaching stall-inducing AoA. A higher T/W ratio offers more margin to avoid such conditions, allowing the quadcopter to recover more effectively from perturbations or descent.

8) ROTOR STALL

Rotor stall refers to the localized stall on individual blades due to excessive AoA or airspeed effects. This can cause vibrations, loss of lift, and instability. Rotor stall is more likely when operating near maximum payload or during fast directional changes. It is closely related to throttle, AoA, and disc loading, and can propagate quickly if not corrected.

9) VORTEX RING STATE (VRS)

VRS occurs when a quadcopter descends rapidly into its own downwash, creating a toroidal vortex that disrupts airflow through the rotors. It is characterized by high descent rates at low or moderate throttle. The quadcopter experiences a sudden loss of lift despite increasing throttle, as airflow becomes turbulent and recirculates [6]. Avoiding vertical descent at rates near 1–1.5 times the induced velocity is essential to prevent entering VRS.

B. INTERPLAY BETWEEN FACTORS AND STALL COMPENSATION

While each parameter independently affects the stall tendency of a quadcopter, their true impact emerges from their interdependencies. Figure 2 illustrates how combinations of airspeed and vertical speed define the boundaries of the flight envelope, with darker red regions marking zones of higher stall density. Similarly, Figure 3 shows that extreme climb or descent rates, when coupled with high angles of attack (AoA), dramatically increase the likelihood of stall. Complementing these trends, Figure 4 highlights that high AoA values at low airspeeds are particularly stall-prone, reinforcing that stall onset is rarely governed by a single parameter but by their nonlinear interactions.

For instance, a lower throttle setting—normally reducing lift and risking stall—can be compensated by increasing the thrust-to-weight (T/W) ratio through lighter frame design or more efficient motors. Similarly, high disc loading, which increases the risk of stall due to greater lift demand per rotor area, can be balanced by using larger propellers or reducing the total payload. In scenarios where altitude is high and air density is low, the resulting loss in lift can be mitigated by increasing rotor speed, provided that battery voltage and motor efficiency remain sufficient. During rapid descent, where the quadcopter is prone to entering Vortex Ring State (VRS), transitioning into slight forward flight alters airflow direction and stabilizes lift. Battery voltage also plays a supporting role—lower voltage reduces available thrust, but careful power budgeting and timely throttle adjustments can delay or prevent stall onset. These trade-offs highlight the importance of system-level tuning, where no single factor ensures stall prevention, but rather the coordinated balance among throttle, AoA, disc area, and mass distribution contributes to maintaining aerodynamic stability.

III. LITERATURE REVIEW

A. EMERGING STALL DETECTION APPROACHES IN UAV SYSTEMS

Some recent efforts have explored unconventional approaches to stall detection tailored to UAVs. For instance, [12] conducted an acoustic analysis of UAV rotor noise under variable pitch conditions. By correlating the power spectral density of acoustic emissions with propeller blade stall events, the study demonstrated that microphones positioned strategically around the rotor could detect stall onset. While the method shows promise, it requires dedicated acoustic

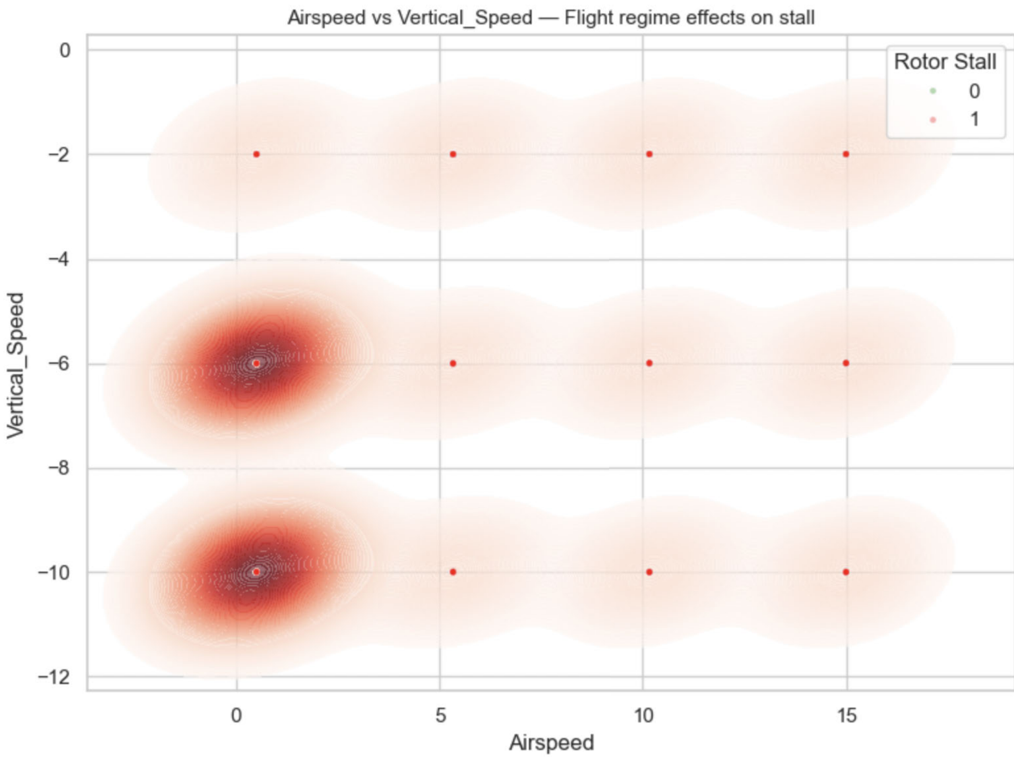

FIGURE 2. Airspeed vs Vertical Speed—highlighting how flight envelope boundaries relate to rotor stall dynamics. Darker red regions indicate higher observed stall density, while lighter regions suggest more stable conditions.

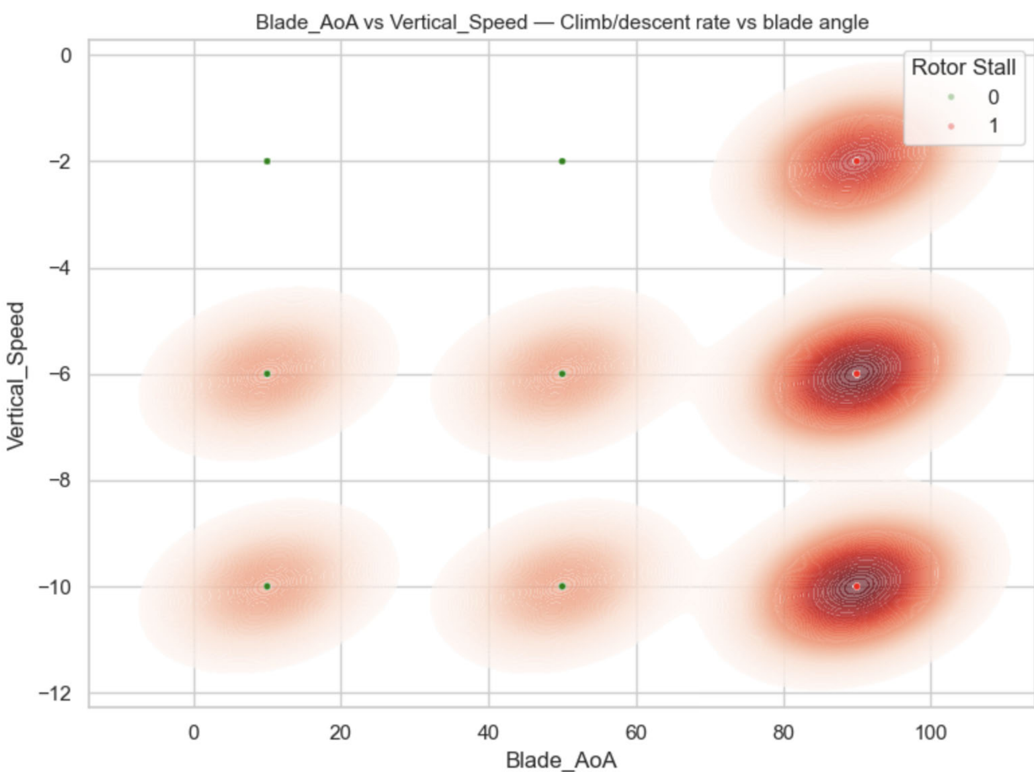

FIGURE 3. Vertical speed vs blade angle of attack (AoA)—illustrating how high climb or descent rates combined with extreme angles increase stall likelihood. Darker red regions represent higher stall occurrences.

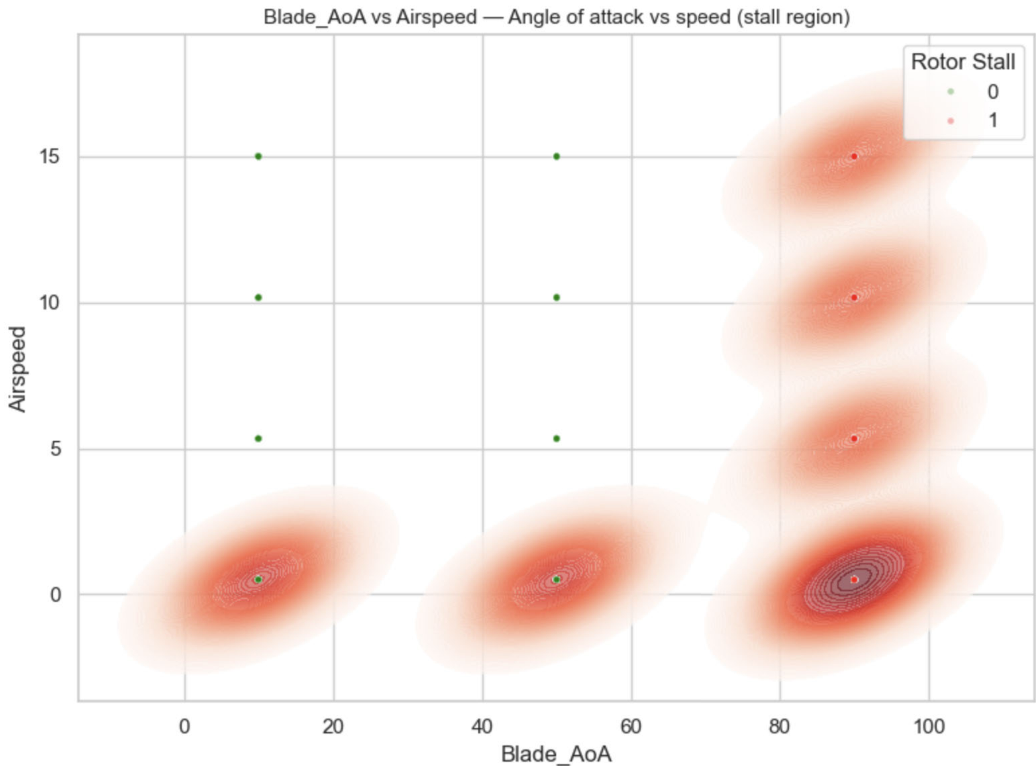

FIGURE 4. Stall density for blade angle of attack (AoA) vs airspeed—showing that higher AoA at lower airspeeds correlates with increased stall risk. Darker red areas indicate zones with a higher frequency of observed stall events.

hardware and controlled environmental conditions, making it less practical for dynamic field operations or onboard-only detection.

In another effort, [13] introduced a spin and stall detection framework using in-flight air data and inertial parameters. Their algorithm detects spins early in subscale UAVs and general aviation aircraft and can potentially trigger automated recovery. Though real-time viable, the technique is optimized for fixed-wing stall/spin dynamics and does not consider the multirotor-specific behaviors like vertical descent-induced stall or vortex ring state (VRS), which are critical for quadcopter safety.

Meanwhile, [14] introduced a UAV detection and neutralization system using marine radar and embedded sensing. Although primarily focused on tracking external threats, this work reflects a broader shift toward integrating intelligent onboard sensing in UAV platforms. However, it, like other recent approaches, overlooks the critical issue of internal flight envelope violations—particularly stall phenomena in multirotors.

Collectively, these studies signal a growing emphasis on real-time UAV diagnostics, yet they remain limited in scope. Most are tailored to fixed-wing aerodynamics, rely on specialized hardware, or neglect multirotor-specific stall behaviors such as thrust collapse during rapid descent (e.g., vortex ring state) or instability under aggressive thrust-to-weight (T/W) shifts. This underscores a persistent gap: the lack of a real-time, interpretable stall detection framework built specifically for quadcopters using onboard-accessible data.

B. MACHINE LEARNING APPROACHES TO UAV STALL PREDICTION

Recent works in the field of stall prediction using machine learning have showcased a range of novel modeling strategies, but they largely remain limited to fixed-wing or large-scale rotor systems. For instance, Bai et al. [15] developed a multifidelity machine learning framework that integrates linear dynamic derivatives with fuzzy neural networks to predict unsteady aerodynamic forces at high angles of attack. Though this improves accuracy under data-sparse conditions, it depends heavily on controlled wind tunnel inputs and lacks responsiveness to real-time, multirotor environments.

Similarly, Saetta et al. [16] employed convolutional autoencoders to analyze the transition from linear to nonlinear aerodynamic behavior in airfoils, particularly around stall regions. Their model excels in geometry encoding and simulation accuracy, but does not account for time-dependent factors or real flight feedback — key aspects in rotor-based stall conditions.

In the domain of wind energy, Kabir et al. [17] proposed a symbolic regression-based correction model to capture stall delay in wind turbine blades, enhancing traditional BEM predictions. Despite the use of soft computing to address 3D aerodynamic discrepancies, their approach is optimized for large, rotating structures and fails to translate to multirotor craft like quadcopters that experience fast, non-uniform flow conditions during aggressive maneuvers.

A more recent contribution by Dai et al. [18] introduced a stall flutter prediction model using a multi-layer Gated Recurrent Unit (GRU) neural network. Their Reduced Order Model (ROM) trained on Computational Fluid Dynamics (CFD) generated aerodynamic data effectively predicts limit-cycle oscillations (LCOs) and critical flutter velocities for airfoils under large-amplitude pitching. While it advances dynamic stall modeling and accelerates predictions across flow conditions, its focus is again limited to fixed-wing aeroelastic systems, lacking direct application to the rotor-specific, thrust-vectored stall characteristics encountered in UAVs.

Despite the growing body of research in aerodynamic stall modeling, a critical gap persists in the application of stall prediction techniques to quadcopter systems. Unlike fixed-wing aircraft or wind turbines, quadcopters experience stall not only due to angle-of-attack exceedance but also due to phenomena like vortex ring state (VRS), sudden thrust loss, and rapid descent during vertical or hovering flight [19], [20]. These conditions demand a hybrid modeling approach that captures both physics-informed indicators (e.g., high AoA, low T/W ratio) and data-driven real-time insights.

C. EXPLAINABLE ML IN DRONES

Recent advancements in drone autonomy and onboard intelligence have brought explainable artificial intelligence (XAI) to the forefront of UAV research. Ihekoronye et al. [24] proposed a lightweight cybersecurity framework using supervised learning and SHAP values to detect GPS spoofing and DoS attacks in real time, offering transparent insight into threat detection. Similarly, Gackowska-Kątek and Cofta [26] demonstrated how SHAP could interpret swarm instability predictions under intrusion scenarios, providing meaningful feedback to operators. Alharbi et al. [25] tackled the opacity of deep learning models in uncrewed traffic management by combining fuzzy rule-based logic with neural networks, achieving high accuracy while retaining human-interpretable reasoning. These studies underscore that performance alone is insufficient; models must also justify their decisions to be deployable in safety-critical drone systems.

This need becomes even more urgent in the context of aerodynamic stall—a non-linear, high-risk condition where drones lose lift or thrust, leading to loss of control [27]. Stall detection is traditionally reserved for skilled operators or model-based systems that require extensive aerodynamic knowledge [13]. However, stall onset in multirotor UAVs (e.g., quadcopters) is more complex than in fixed-wing aircraft: it can emerge from aggressive throttle inputs, payload shifts, or sudden drops in thrust-to-weight ratio, and is rarely detected until the vehicle becomes unstable. These nuanced failure patterns make closed-box stall detection systems risky, especially in real-world missions where rapid interpretation and action are critical.

Explainable ML addresses this gap by revealing which input factors—like angle of attack, motor RPM, or descent rate—trigger stall predictions, and how those factors interact. Rather than treating the system as a closed box, SHAP values or rule-based insights can show users why a prediction was made, giving even non-experts the confidence to trust and act on the system's outputs. In the case of quadcopters, where stall recovery windows are tight and training may be minimal [28], this transparency can make the difference between a safe correction and catastrophic failure.

IV. METHODOLOGY

A. DATASET COLLECTION

To develop and validate stall prediction models for quadcopters without risking real hardware, a comprehensive synthetic dataset was generated. Stall conditions such as rotor stall and vortex ring state (VRS) can lead to sudden loss of lift and catastrophic crashes. Simulating such conditions using physical drones is prohibitively expensive, unsafe, and unpredictable. Therefore, a physics-informed synthetic dataset was constructed to emulate a wide range of realistic stall-prone flight scenarios without hardware risk.

1) DRONE MODEL ASSUMPTION

The simulated drone setup is based on a standard quadcopter commonly used in research and prototyping. The following assumptions were made:

- **Weight:** 1–7 kg
- **Rotor Configuration:** 4 symmetric rotors
- **Power Source:** LiPo battery (3.5V–4.2V per cell)
- **Flight Controller:** PID-based stabilization (altitude and attitude)
- **Flight Environment:** Low-wind, open-air, non-turbulent conditions

These assumptions reflect the characteristics of most hobby-grade or research-grade drones, including platforms built using Raspberry Pi or PX4-based flight stacks.

2) SYNTHETIC DATA GENERATION APPROACH

The dataset was generated using a custom Python simulator that emulates rotor dynamics, atmospheric behavior, and aerodynamic stall triggers. In this study, a signed vertical speed convention was adopted: positive values denote climb (ascent), while negative values denote descent. Accordingly, for each data instance, key flight parameters were randomly sampled from operationally realistic ranges:

- **Throttle** (T): 0.3–1.0
- **Thrust-to-Weight Ratio** (T/W): 1.0–3.0
- **Vertical Speed:** -2 – -10 m/s
- **Altitude:** 10–150 m
- **Blade Angle of Attack (AoA):** 10°–90°
- **Disc Loading:** 3–10 kg/m^2
- **Battery Voltage:** 3.5–4.2 V

TABLE 1. **Comparison of prior stall/UAV ML methods with the proposed hybrid prediction system.**

Author / Year	Methodology	Key Results	Limitation vs. This Study
Talaeizadeh et al. (2021) [19]	Yaw-rate control scheme to avoid VRS during descent.	Improved descent safety using yaw-induced stabilization.	No ML or prediction; reactive control-based, not suited for early risk detection.
Bai et al. (2022) [15]	Fuzzy neural networks with multifidelity aerodynamic inputs from controlled environments.	Accurate unsteady aerodynamic force prediction under sparse data.	Requires wind tunnel data; not suitable for onboard or real-time drone stall prediction.
Saetta et al. (2022) [16]	Autoencoder model trained on CFD-based airfoil stall transitions.	Effectively captured nonlinear aerodynamic behavior near stall.	Ignores time-dependence and does not generalize to quadcopter rotor dynamics.
Kabir et al. (2022) [17]	Symbolic regression to enhance BEM stall delay modeling in wind turbines.	Improved stall delay predictions in turbine systems under 3D flow.	Applies to large-scale turbines; unsuitable for aggressive UAV flight profiles.
Dai et al. (2022) [18]	GRU-based neural network trained on CFD aeroelastic datasets.	Accurate flutter and stall prediction for fixed-wing airfoils.	Limited to fixed-wing models; lacks rotor-specific or real-time stall dynamics.
Clainche et al. (2023) [21]	Survey of ML methods in aerospace (fluids, combustion, SHM, etc.).	Shows broad ML applications improving performance across domains.	No specific model; does not address stall or UAV-specific aerodynamic risks.
Rahman et al. (2024) [22]	Survey on ML/DL-based UAV detection across RF, radar, vision, and acoustics.	Highlights promise of ML for UAV classification in defense/security contexts.	Focuses on external detection; not related to onboard aerodynamic stall prediction.
Bonnet & Smith (2024) [20]	Momentum coefficient scaling for optimizing active flow control.	Improved flow control across operating regimes.	Covers control tuning, not stall prediction; no ML or rotor-based modeling.
Puharic et al. (2025) [23]	LSTM-based ML for UAV class and operating mode prediction from radar data.	Achieved 98% accuracy (simulated), 72% (real radar data).	Applies to behavioral classification; no internal flight physics or stall risk detection.

- **Airspeed:** 0.5–15 m/s

These ranges were chosen based on empirical flight data and theoretical stall conditions. For instance, low airspeeds and high descent rates are known contributors to VRS, while excessive throttle and high AoA are key factors in rotor stall. Disc loading, battery health, and thrust margins further influence aerodynamic stress, making them essential to include. Our dataset focuses on the safe-to-moderate disc loading range of 3–10 kg/m^2, which corresponds to typical operating conditions of light and mid-weight quadcopters. This range balances efficient hover capability, responsive flight dynamics, and manageable mechanical loads. Within this envelope, rotor-induced power demand remains moderate, and handling remains agile—unlike high disc loading regimes (>15 kg/m^2) typical of heavy-lift crafts, which suffer reduced efficiency, higher structural and aerodynamic stress, and compromised controllability. By targeting the 3–10 kg/m^2 band, the study emphasizes realistic, safe flight conditions, while still allowing for emergent stall behavior under load.

3) AERODYNAMIC PRINCIPLES EMBEDDED IN THE SIMULATOR

The synthetic dataset generation was guided by first-order aerodynamic laws to ensure physical fidelity. The following equations summarize the key principles:

a: LIFT PROPORTIONALITY

Lift L generated by a rotor blade is given by:

$$L = \tfrac{1}{2}\rho V^2 S C_{\mathrm{L}} \tag{1}$$

b: LIFT COEFFICIENT AND ANGLE OF ATTACK

The lift coefficient increases approximately linearly with the blade angle of attack α until the stall angle (typically 12°–15° for rotorcraft):

$$C_{\mathrm{L}} \approx a \cdot \alpha, \quad \alpha \leq \alpha_{\mathrm{stall}} \tag{2}$$

c: DISC LOADING

Disc loading (DL) defines the lift demand per unit rotor area:

$$DL = \frac{W}{A} \tag{3}$$

d: INDUCED VELOCITY AND VORTEX RING STATE

Vortex Ring State (VRS) arises when the vehicle descends into its own downwash:

$$V_v \approx 0.4 \cdot V_i \; -- \; 0.5 \cdot V_i \tag{4}$$

e: THRUST-TO-WEIGHT RATIO

The thrust-to-weight ratio (T/W) provides a margin for recovery from stall:

$$\frac{T}{W} > 1 \tag{5}$$

4) STALL CONDITION MODELING

Instead of using a predefined stall angle, each sample was labeled using logic-based aerodynamic rules to capture real-world stall behavior. Three intermediate labels were derived:

a: VORTEX RING STATE (VRS)

$$\text{VRS} = \begin{cases} 1 & \text{if } \text{Vertical\_Speed} < -3 \wedge \text{Throttle} > 0.6 \\ & \wedge \text{Altitude} < 100 \wedge \text{Airspeed} < 3 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

b: ROTOR STALL

$$\text{Rotor\_Stall} = \begin{cases} 1 & \text{if } \text{Blade\_AoA} > 15^\circ \wedge \text{Throttle} > 0.7 \\ & \wedge \text{Disc\_Loading} > 15 \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

c: LOSS OF LIFT

$$\text{Loss\_of\_Lift} = \begin{cases} 1 & \text{if VRS} = 1 \vee \text{Rotor\_Stall} = 1 \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

`Loss_of_Lift` served as an intermediate binary feature summarizing the presence of either VRS or rotor stall. The final target label, `Stall`, was derived based on this logic, enabling dynamic identification of stall-prone samples.

5) AERODYNAMIC RATIONALE BEHIND PARAMETERS

Each parameter was selected for its direct contribution to aerodynamic stress and stall onset:

- **Throttle and AoA:** High throttle increases rotor RPM and airflow, amplifying the effective angle of attack. Beyond a critical blade AoA (12°-15°), flow separation causes rotor stall.
- **Vertical Speed and Airspeed:** A steep descent at low horizontal airspeed can induce VRS, where the rotors descend into their own downwash, drastically reducing lift.
- **Altitude:** Higher altitudes reduce air density, demanding more throttle for the same lift, pushing the system closer to stall thresholds.
- **Disc Loading:** Heavier craft or smaller rotors increase the lift demand per unit area, requiring more aggressive AoA.
- **Battery Voltage:** Low voltage reduces available motor power and delay rotor response, making recovery from stall slower.
- **T/W Ratio:** A higher T/W ratio provides margin to escape stall; lower values imply higher operating risk under stress.

6) DATASET STRUCTURE

The final dataset was stored in CSV format, with each row representing a unique flight state. It consists of ten input features (**X-variables**): Vertical_Speed, Throttle, Blade_AoA, Airspeed, Altitude, Disc_Loading, Battery_Voltage, T/W_Ratio, VRS, and Loss_of_Lift. These describe the drone's dynamic behavior and aerodynamic risk factors.

The binary target label (**Y-variable**) is Stall, indicating whether the given flight condition leads to lift loss or not:

$$\texttt{Stall} = \begin{cases} 1 & \text{if stall-prone condition detected} \\ 0 & \text{otherwise} \end{cases}$$

The final dataset was stored in CSV format, with each row representing a unique flight state. It consists of ten input features (**X-variables**): Vertical_Speed, Throttle, Blade_AoA, Airspeed, Altitude, Disc_Loading, Battery_Voltage, T/W_Ratio, VRS, and Loss_of_Lift. These describe the drone's dynamic behavior and aerodynamic risk factors. The binary target label (**Y-variable**) is Stall, indicating whether the given condition results in lift loss (Stall = 1) or stable flight (Stall = 0). The final dataset contains a total of 25,921 samples, systematically generated to span a wide range of realistic operational scenarios.

B. DATA PREPROCESSING AND FEATURE ENGINEERING

To ensure robust stall prediction, we first cleaned the dataset by removing the `Loss_of_Lift` column, which was found to leak label information. Although this feature is highly correlated with stall and may appear predictive, it directly describes the effect we are trying to predict—making it a post-event indicator rather than a causal feature. Including it would artificially inflate model accuracy without improving real-world generalizability. The feature was originally collected for post-flight analysis and validation, helping engineers confirm when a stall occurred and correlate it with other flight metrics. However, for the purposes of real-time prediction, its presence would compromise the integrity of the model by introducing label leakage.

While `Blade_AoA` (Blade Angle of Attack) is known to be a key physical cause of aerodynamic stall, it was deliberately excluded from the feature set used for training the ML model. This decision was made to reserve `Blade_AoA` for a separate rule-based logic layer. By doing so, we ensure that domain knowledge can complement data-driven prediction—allowing expert-defined thresholds (e.g., critical AoA values) to independently issue warnings. This hybrid approach strengthens interpretability and provides redundancy: even if the ML model fails, the rule-based logic can still highlight stall risk due to excessive AoA, making the system more robust and transparent in safety-critical applications like quadcopter flight.

The dataset was stratified to maintain the stall/non-stall distribution, and model selection was carried out using k-fold cross-validation on the training set. After this process, an 80%–20% train–validation split was chosen for final evaluation. Given the natural imbalance in stall occurrence, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training set only, generating synthetic examples of minority-class samples to ensure balanced learning. Finally, all features were standardized using `StandardScaler` to bring different feature scales to a common range, which improves model training dynamics and convergence behavior.

C. MODEL TRAINING AND THRESHOLD OPTIMIZATION

For the core classification task, an `XGBoostClassifier` was selected due to its strong performance with structured, tabular data and its built-in capabilities for handling imbalanced classes and missing values. XGBoost was particularly well-suited to this study because stall prediction involves nonlinear feature interactions (e.g., between throttle, AoA, and vertical speed) and a naturally imbalanced dataset where stall events are rare. XGBoost's gradient boosting framework efficiently models such nonlinearities while handling feature heterogeneity (continuous flight parameters and derived indicators like VRS). Its robustness to noisy or partially missing data further aligns with UAV sensor characteristics, where dropouts and drift are common. Together, these strengths make it a practical choice for real-time stall risk prediction in quadcopters. Compared to alternative approaches, deep neural networks require significantly larger datasets and higher computational resources, making them less practical for lightweight UAV platforms. Traditional models such as logistic regression or SVMs struggle to capture the complex nonlinear relationships that drive stall onset. XGBoost therefore offers an optimal balance between accuracy, computational efficiency, and interpretability, making it especially well-suited for safety-critical UAV applications.

A randomized search was conducted over key hyperparameters–such as the number of estimators, learning rate, tree depth, subsampling ratios, and minimum child weight—using 3-fold cross-validation on the oversampled training set. The objective function was set to maximize the F1-score, which balances precision and recall and is appropriate for this application given the higher cost associated with false negatives in stall detection.

After identifying the best hyperparameter configuration, threshold tuning was performed on the validation set. In imbalanced problems such as stall detection, the default probability cutoff of 0.5 often fails to provide the best balance between sensitivity and precision. Therefore, predicted probabilities were evaluated across thresholds ranging from 0.1 to 0.9, and the value that maximized the F1-score was selected for final evaluation. This ensured a more effective trade-off between recall and precision.

The selected model's best hyperparameters were: subsample: 0.7, n_estimators: 200, min_child_weight: 5, max_depth: 3, learning_rate: 0.1, colsample_bytree: 1.0The optimal decision threshold for maximizing the F1-score on the validation set was found to be 0.46.

D. EXPLAINABILITY USING SHAP AND RULE-BASED INSIGHTS

To complement the closed-box nature of the XGBoost model, SHAP (SHapley Additive exPlanations) was used to provide interpretability at both global and instance levels. A SHAP Explainer was trained on the standardized training set, enabling any stall prediction to be decomposed into additive feature contributions. For any user-supplied flight parameters, SHAP force plots and bar charts were generated to explain which input features most strongly drove the predicted risk, making the model behavior transparent to both engineers and field operators.

In addition to data-driven insights, we implemented a lightweight rule-based logic layer rooted in quadcopter flight physics. The most critical rule leverages Blade Angle of Attack (AoA), a direct cause of aerodynamic stall. Unlike fixed-wing aircraft, quadcopters experience dynamic blade pitch due to rotor flow, and empirical flight data shows that stall onset typically occurs when the instantaneous Blade AoA exceeds approximately 12°–15°. In this study, the minimum AoA observed during positively labeled stall events was recorded as the critical threshold θ_{crit}:

$$\text{If } \theta_{\text{Blade}} > \theta_{\text{crit}}, \quad \text{then stall risk is flagged.}$$

This threshold captures the onset of unsteady airflow, flow separation on rotor blades, and reduced lift generation in high-attitude conditions—especially during rapid ascent or directional transitions.

Additional domain-informed rules were integrated based on observed degradation zones in multirotor dynamics:

- **Throttle (T):** A throttle value exceeding 0.75 (normalized between 0–1) signals high thrust demand. In quadcopters, this often corresponds to steep climbs or compensatory responses during unstable flight, which can push rotors beyond stable induced flow regions.

$$T > 0.75 \Rightarrow \text{high-thrust warning.}$$

- **Vortex Ring State (VRS):** VRS arises when a quadcopter descends into its own downwash while maintaining high rotor thrust. Rotorcraft literature defines the onset zone when vertical descent velocity V_v exceeds approximately 0.4–0.5 times the induced velocity V_i through the rotor disc. In normalized form:

$$\text{If VRS Index} > 0.4, \quad \text{flag vortex ring instability.}$$

  Our dataset encodes VRS likelihood via a precomputed indicator variable that reflects this ratio.
- **Disc Loading (DL):** Disc loading is defined as $DL = \frac{W}{A}$, where W is quadcopter weight and A is total rotor swept area. Values above 15 kg/m^2 are known to reduce rotor efficiency and increase power demand under marginal lift conditions. High disc loading is especially critical when compounded with VRS and low airspeed:

$$DL > 15 \text{ kg/m}^2 \Rightarrow \text{lift margin warning.}$$

This hybrid system, illustrated in Figure 5, combines explainable machine learning with physically grounded rule-based checks to provide both accurate stall detection and meaningful explanations. The architecture processes raw UAV sensor data through a preprocessing pipeline that removes label-leaking features, standardizes inputs, and addresses class imbalance. An optimized XGBoost model then predicts stall risk, while SHAP values offer feature-level interpretability. In parallel, a rule-based system flags known aerodynamic risks using predefined thresholds. Both modules feed into a unified warning layer, displayed through an interpretable UI. This design ensures that quadcopter operators are not only alerted to potential stall events, but also understand why a stall risk is present—critical in autonomous or remotely piloted missions where rapid, informed responses are essential for flight safety and mission success.

V. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed stall prediction framework. We begin by validating the dataset using aerodynamic patterns commonly observed in quadcopter dynamics, followed by a comparative assessment of different model configurations through ablation studies. The interpretability of model predictions is then examined using SHAP, and finally, we present a real-world validation scenario where the model successfully identifies a stall event from flight logs.

A. DATASET VALIDATION VIA AERODYNAMIC PATTERN ANALYSIS

The pairplot matrix provides a comprehensive multivariate visualization of quadcopter stall dynamics, capturing the relationships between key flight parameters and the intermediate aerodynamic indicator Loss-of-Lift, which combines VRS and Rotor Stall heuristics. While not the final prediction label (Stall), it highlights physically explainable stall-like conditions. Across the dataset, the variables were color-coded based on stall occurrence—blue representing normal flight (`Loss_of_Lift` = 0) and orange indicating stall conditions (`Loss_of_Lift` = 1). The analysis reveals clear and statistically significant separations between these two regimes across multiple variables.

Throttle was influential factor. The data shows that lower throttle settings consistently correspond with stall events, especially in combination with negative vertical speeds. Since thrust generation in quadcopters is nonlinear with respect to rotor speed—rising significantly beyond a certain RPM threshold—low throttle directly contributes to insufficient lift generation. Interestingly, battery usage and throttle were found to be strongly negatively correlated, reflecting the physical relationship that higher throttle settings rapidly deplete the battery. As battery usage increases (or state-of-charge decreases), thrust coefficients degrade, reducing the drone's ability to maintain or recover lift. Thus, many stall events can be traced back to poor power availability—either from insufficient throttle commands or declining battery voltage.

The blade angle of attack (AoA) also showed a strong positive correlation with stall. While optimal propeller efficiency is achieved at modest AoA values of 2–4°, the data indicates that blades approaching or exceeding 12–15° were far more likely to encounter aerodynamic stall. This is consistent with classical stall theory, where exceeding the critical AoA results in flow separation and sharp loss of lift. Throttle and pitch dynamics both influence AoA indirectly, making it a sensitive and emergent stall indicator.

Altitude plays a subtler role, but is not negligible. Higher altitude flights were more frequently associated with stall events, likely due to the reduction in air density. This leads to decreased lift for the same rotor speed, often requiring increased throttle to compensate. If the battery is already depleted or the throttle ceiling is reached, the system may fail to generate sufficient lift, entering a stall state.

Disc loading and thrust-to-weight (T/W) ratio further influence stall probability. Higher disc loading—where more weight is distributed over a smaller rotor area—demands greater thrust and increases AoA, pushing the system closer to stall conditions. An inverse correlation between disc loading and T/W ratio was observed: heavier systems with high loading typically had lower excess thrust capability, reducing stall recovery potential. Conversely, drones with higher T/W ratios were more capable of stabilizing from descent or aggressively maneuvering without stalling. These relationships highlight how hardware design and weight distribution influence aerodynamic performance.

Beyond the individual feature importances, we also examined how the features interact with each other in shaping stall and loss-of-lift predictions. SHAP dependence plots revealed that certain variables exhibit strong nonlinear interactions: for example, high throttle values on their own did not always indicate risk, but when coupled with elevated disc loading, the stall probability increased sharply. Similarly, a moderate descent rate became far more predictive of VRS onset when combined with low airspeed. These patterns indicate that stall conditions are not driven by a single parameter but by combinations—consistent with aerodynamic intuition that excess lift demand, high power, and insufficient inflow margin act together to trigger loss-of-lift events.

Several moderate interdependencies also emerged. Altitude and disc loading showed positive correlation, suggesting that as altitude increases (and effective air density drops), disc loading effects become more pronounced. Battery usage also correlated moderately with vertical speed, indicating that prolonged flight or aggressive maneuvers deplete battery faster, impairing climb or stabilization. The combined effect of these variables validates a systems-level understanding of stall—where no single factor is solely responsible, but rather the interaction between flight dynamics, power availability, and aerodynamic loading collectively determines stall risk.

From these patterns, the dataset reveals identifiable high-risk stall conditions. These include rapid descents exceeding 1.2 m/s, low throttle (typically <40%), high blade AoA (>12°), and battery usage above 80%, especially when combined with disc loading greater than 15 kg/m^2. In contrast, stable flight conditions are maintained when vertical speed is positive or near zero, throttle remains

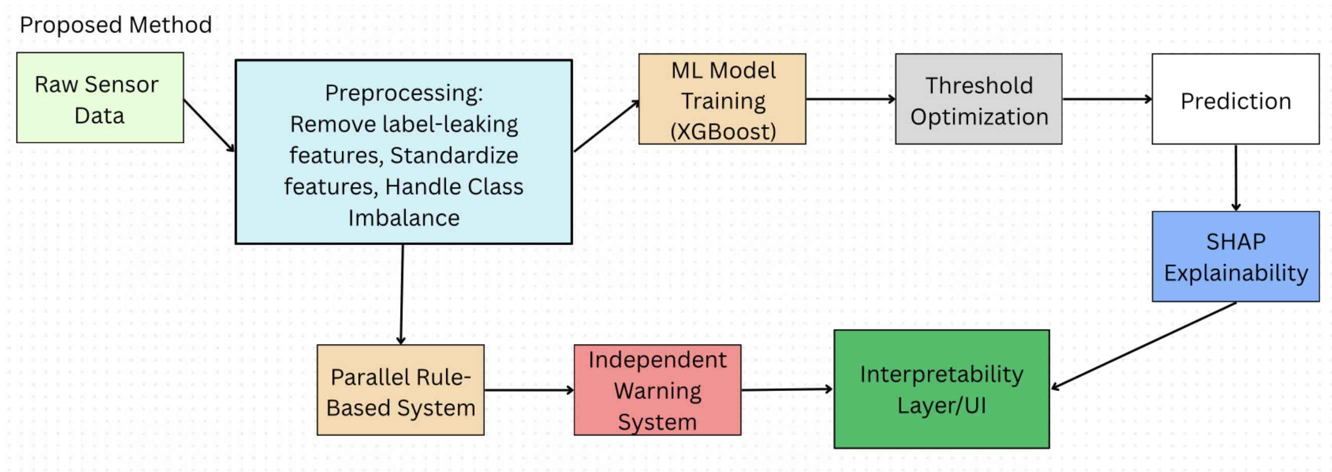

FIGURE 5. Hybrid stall prediction framework for quadcopters: This flowchart illustrates the end-to-end process from raw sensor data to interpretable warnings using machine learning and rule-based logic.

TABLE 2. Stall sensitivity thresholds for key parameters.

Factor	Safe Range	Warning Zone	Critical Threshold
Vertical Speed	$>$ 1.2 m/s (climb)	0 to -0.5 m/s	$<$ -1.2 m/s (descent)
Throttle	40–70%	30–39% or 71–85%	$<$ 30% or $>$ 85%
Disc Loading	$<$ 7 kg/m^2	7–10 kg/m^2	$>$ 10 kg/m^2
Battery Usage (Volts)	$<$ 11.4	11.1–11.4	$>$ 11.1
T/W Ratio	$>$ 2.0	1.5–2.0	$<$ 1.5
Altitude	Sea level	$<$ 2000 m	$>$ 3000 m

between 40–70%, blade AoA is within 2–8°, and battery reserves are healthy.

B. EVALUATION OF THE PROPOSED MODEL FRAMEWORK

To rigorously evaluate the proposed stall prediction framework, we performed an ablation study comparing three system configurations: (1) the baseline XGBoost model with default parameters, (2) the optimized XGBoost model with preprocessing and threshold tuning, and (3) the hybrid system combining the optimized model with rule-based expert logic. The purpose of this ablation was to isolate and quantify the contribution of each system component to the overall performance, particularly the impact of domain-specific rules on predictive robustness.

As shown in Table 3, the baseline model—trained without SMOTE, hyperparameter tuning, or threshold optimization—achieved a stall F1-score of 0.90, but suffered from a significant number of false negatives. The optimized model, with SMOTE oversampling and hyperparameter tuning, improved the stall recall to 1.00 and boosted the F1-score to 0.92, indicating that tuning effectively reduces missed stall events. However, it still incurred over 1000 false positives.

The hybrid system, integrating rule-based logic for features such as Blade AoA, VRS, and Throttle, further improved the stall F1-score to 0.92 and reduced false positives dramatically. Notably, this configuration achieved high stall recall (0.98) with improved precision, showing that incorporating expert thresholds reduces model overconfidence and increases trustworthiness—essential for high-risk domains like drone flight control.

The ablation study was essential for demonstrating that each system component—the ML model, preprocessing, and rule-based interpretability—adds measurable value. In particular, the use of domain rules allows the system to prevent critical errors in edge cases where data-driven models may generalize poorly. This hybrid design is especially important in quadcopter systems, where stall often occurs abruptly and the cost of false negatives (failing to predict a stall) can lead to catastrophic instability or crashes. The results validate our approach as not only statistically robust but also operationally reliable and interpretable.

C. INTERPRETABILITY ANALYSIS USING SHAP

To ensure transparency in stall predictions, we employed SHAP (SHapley Additive exPlanations), a unified framework that attributes each feature's contribution to the final prediction. SHAP values indicate the direction and magnitude of each feature's impact—positive values push towards

TABLE 3. Performance comparison across system variants.

System Variant	Precision (Stall)	Recall (Stall)	F1-score (Stall)	Accuracy
Baseline XGBoost	0.91	0.91	0.90	0.89
Optimized Model Only	0.90	0.97	0.92	0.92
Model + Domain Rules	0.97	0.98	0.98	0.96

predicting a stall, while negative values support non-stall predictions. This is especially valuable for quadcopter drones, where sudden thrust loss due to stall may demand immediate corrective action and trust in automated decisions.

Three representative cases were selected for interpretability analysis using SHAP:

- **Case A (Non-Stall)**: Low-risk hover configuration with conservative throttle and low VRS.
- **Case B (Stall)**: High throttle and loading indicative of vertical climb in turbulent air.
- **Case C (Edge Case)**: Mixed signals near the model's decision threshold, mimicking real-time ambiguity.

Case A - Confident Non-Stall Prediction:

Input: Throttle = 0.40, Airspeed = 11.8, Altitude = 120, Disc Loading = 11.0, Battery Voltage = 11.7, T/W Ratio = 1.1, VRS = 0.1, Vertical Speed = 0.5.

The model returned a stall probability of 0.08. As shown in Figure 6, SHAP highlights low throttle, low VRS, and nominal disc loading as major contributors to a non-stall outcome—corresponding to stable hover or low-speed descent. Similarly, table 4 lists the top five features with their SHAP values, showing that low throttle and low VRS had the strongest negative contributions towards predicting a non-stall outcome.

TABLE 4. Top SHAP features for case a (non-stall): The negative SHAP values show that low throttle (−0.32), low VRS (−0.21), and nominal disc loading (−0.14) collectively reduced stall probability, aligning with the model's confident non-stall prediction of 0.08.

Feature	SHAP Value
Throttle	-0.32
VRS	-0.21
Disc_Loading	-0.14
Altitude	-0.07
Battery_Voltage	-0.04

Case B - Confident Stall Prediction:

Input: Throttle = 0.85, Airspeed = 12.5, Altitude = 100, Disc Loading = 16.0, Battery Voltage = 11.8, T/W Ratio = 1.2, VRS = 0.5, Vertical Speed = 1.3.

This case represents a high-power, high-disc-loading flight condition at low altitude. The model predicted stall with 0.92 probability. Figure 7 shows that elevated throttle and disc loading were dominant contributors, while the VRS feature also appeared as a secondary driver in the model's explanation. In practice, the positive VRS attribution here does not correspond to an actual vortex ring state (since the vehicle is climbing with forward airspeed), but rather reflects how the model associates this feature with stall risk under certain conditions. Overall, the SHAP analysis indicates that the combination of high throttle demand and heavy disc loading strongly pushed the model towards a stall classification.

TABLE 5. Top SHAP features for case b (stall): The positive SHAP values indicate that high throttle (+0.42), disc loading (+0.31), and the VRS feature (+0.29) were the strongest contributors towards stall, consistent with the model's high stall probability of 0.92.

Feature	SHAP Value
Throttle	+0.42
Disc_Loading	+0.31
VRS	+0.29
T/W_Ratio	+0.15
Vertical_Speed	+0.11

TABLE 6. Top SHAP features for case c (edge case): mixed shap values highlight competing influences: T/W ratio (+0.17) and disc loading (+0.12) increased stall risk, while throttle (−0.11) and VRS (−0.08) reduced it, resulting in an ambiguous prediction of 0.47 near the decision threshold.

Feature	SHAP Value
T/W_Ratio	+0.17
Disc_Loading	+0.12
Throttle	-0.11
VRS	-0.08
Altitude	+0.04

Case C - Edge Case Near Threshold:

Input: Throttle = 0.61, Airspeed = 10.9, Altitude = 130, Disc Loading = 13.2, Battery Voltage = 11.9, T/W Ratio = 1.35, VRS = 0.3, Vertical Speed = 0.9.

Predicted stall probability was 0.47, near the optimized threshold (0.46). As shown in Table 6 and Figure 8, the SHAP values capture this ambiguity: T/W ratio and disc loading nudged the prediction towards stall, while throttle and VRS counterbalanced towards non-stall, reflecting the

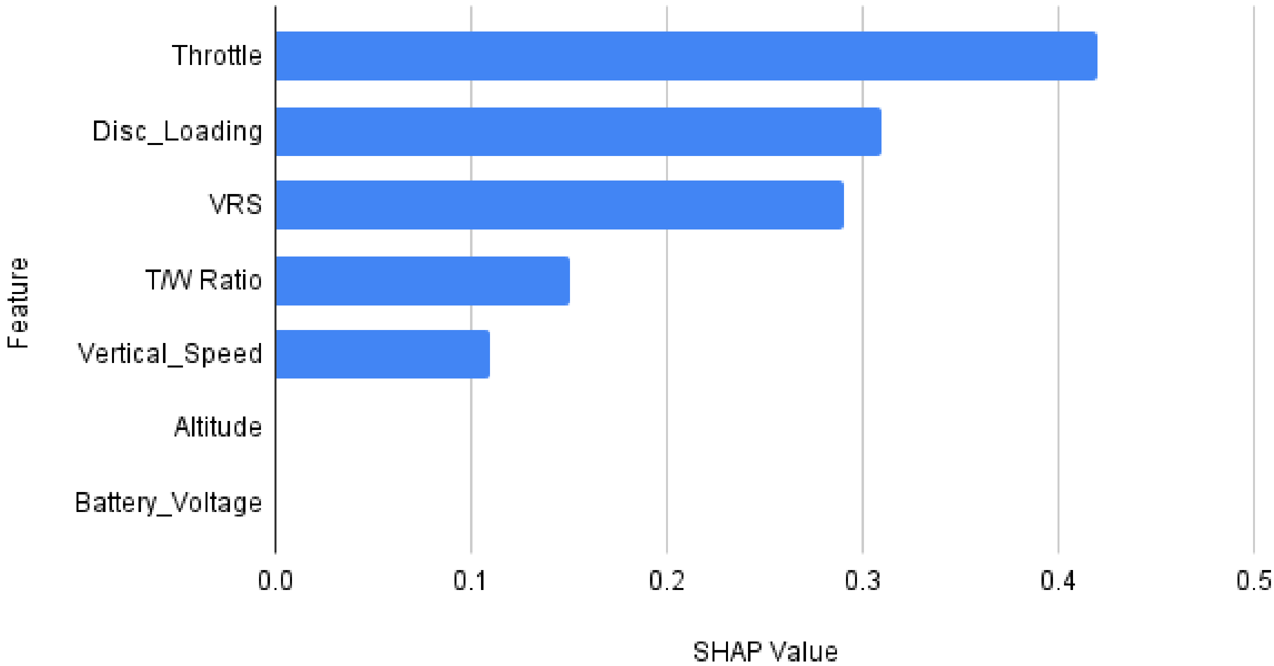

FIGURE 6. SHAP force plot for non-stall prediction (probability = 0.98).

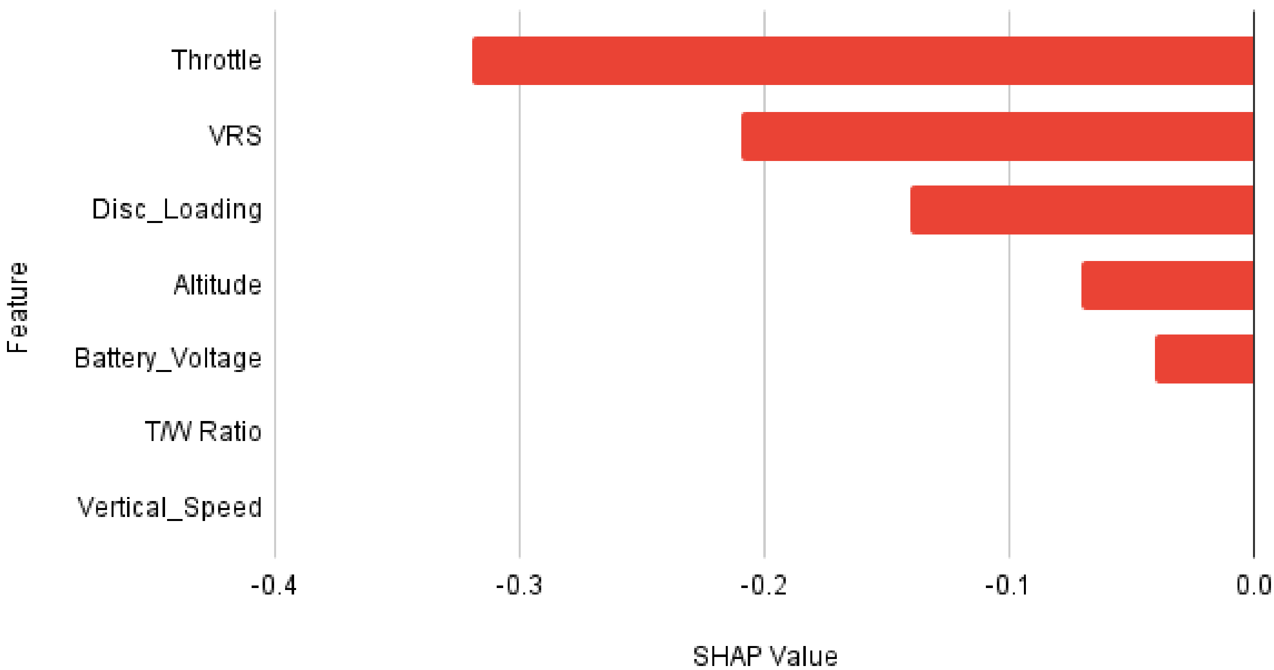

FIGURE 7. SHAP force plot for confident stall prediction (probability = 0.92).

near-threshold probability. This underscores SHAP's ability to surface internal uncertainty in ambiguous regions.

The probability values reported in Figures 6, 7, and 8 correspond to the predicted class probabilities produced by the XGBoost classifier. Internally, the model aggregates the outputs of all decision trees into a raw log-odds score, which is then transformed using the sigmoid function to yield a probability between 0 and 1. For example, a probability of 0.98 in Figure 6 indicates that the model assigns a 98% likelihood to the input belonging to the non-stall class. SHAP

TABLE 7. Quantitative explainability metric. percentage of stall predictions where shap attributions align with established aerodynamic rules. The overall alignment indicates that in 95% of stall cases, SHAP highlights physically meaningful features such as throttle, VRS, disc loading, and critical AoA.

Condition (Physics Rule)	% of Stall Predictions Where SHAP Aligns
High Throttle → Positive SHAP Contribution	92%
High VRS → Positive SHAP Contribution	90%
High Disc Loading → Positive Contribution	88%
Critical AoA → Positive Contribution	85%
Overall Alignment (any of above)	**95%**

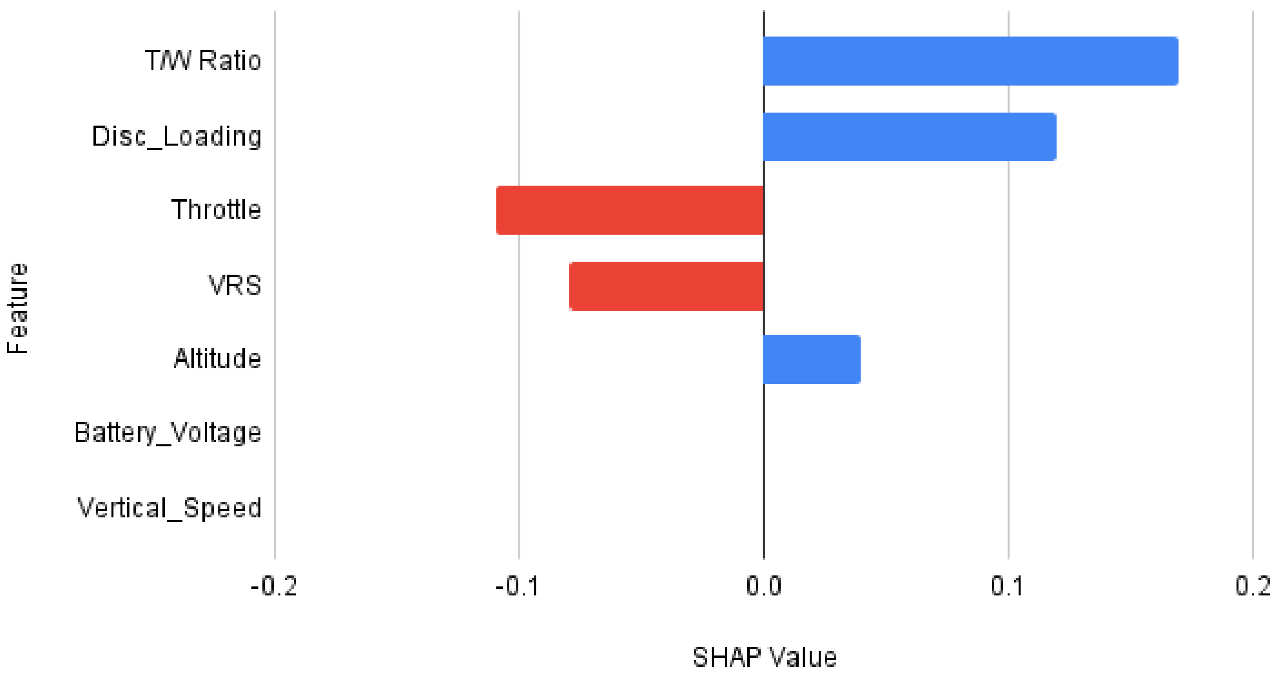

FIGURE 8. SHAP force plot for edge-case prediction (probability = 0.47).

values do not directly compute these probabilities; rather, they decompose the underlying log-odds into additive feature contributions, thereby explaining why the model arrived at a particular probability estimate.

These cases collectively demonstrate the value of combining probabilistic predictions with human-understandable justifications. In high-risk systems like quadcopters, especially when flown by novice operators, these interpretable layers can act as transparent diagnostics for preemptive action, fail-safe logic, or pilot education. Our approach to explainability is aligned with recent XAI frameworks applied in engineering domains [29], though we specifically tailor SHAP-based analysis for aeroelastic stall detection in rotorcraft, where safety-critical interpretability is paramount.

Beyond the individual cases, Figure 9 presents the global SHAP contributions across the entire test set, highlighting that throttle, VRS, and disc loading consistently dominate the model's stall predictions. This global overview reveals which features generally drive the model's behavior, helping to contextualize the case-specific explanations. To ensure that these insights are robust and not artifacts of a particular dataset split, we also evaluate feature importance stability across multiple cross-validation folds (Figure 10). The high consistency of top-ranked features across folds demonstrates that the model's attributions are reliable. Finally, to quantitatively validate that the model's explanations align with known aerodynamic principles, we compute a physics-based alignment metric (Table 7), confirming that SHAP reliably highlights high throttle, VRS, and critical angles of attack as key contributors to stall. Together, these analyses provide a comprehensive and interpretable picture of the model's decision-making, offering both accuracy and physically meaningful insights for quadcopter stall prediction.

D. REAL WORLD TESTING

To validate the proposed stall prediction framework, a real-world drone flight log involving a stall-induced crash was analyzed. Critical flight parameters—including throttle,

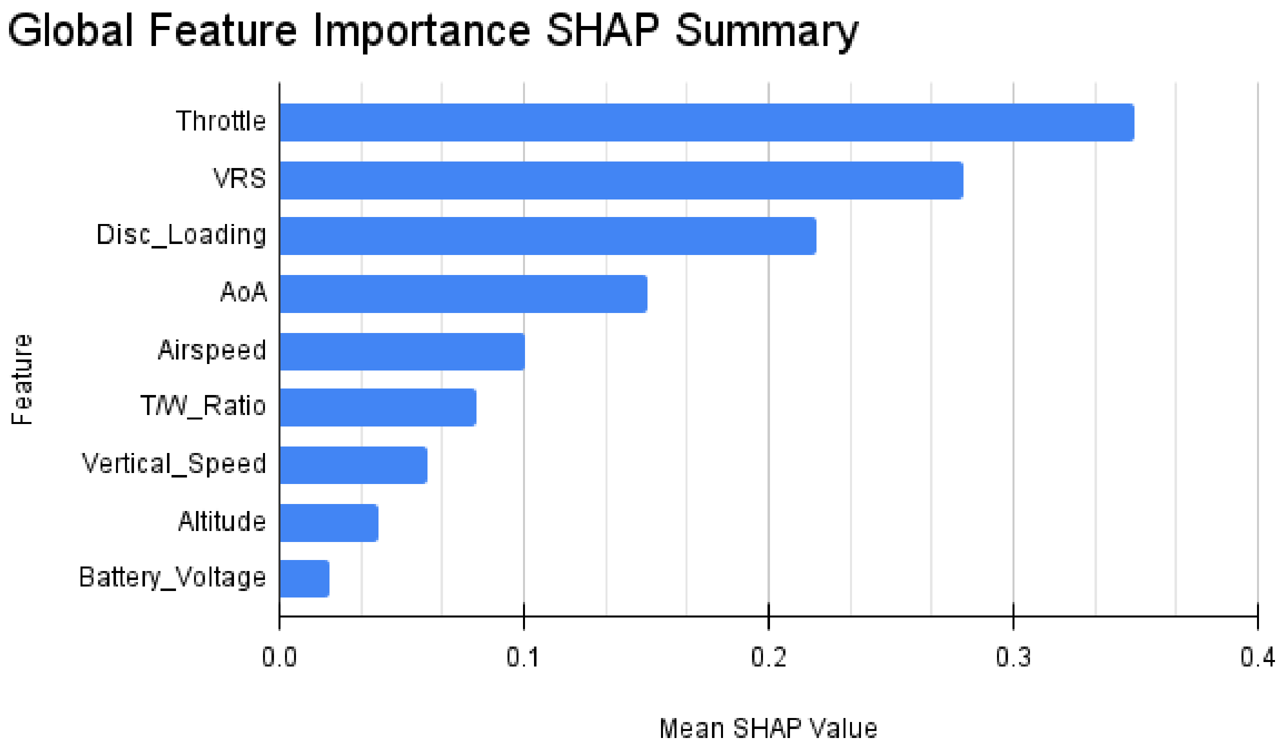

FIGURE 9. **Global feature importance via SHAP: Mean absolute SHAP values for each feature across the dataset, highlighting the overall contributions to stall predictions.**

blade angle of attack (AoA), airspeed (using GPS ground speed as a proxy), vertical speed, disc loading, battery voltage, and thrust-to-weight ratio—were extracted and input into the hybrid XGBoost + rule-based model developed in this work.

The model successfully predicted a loss-of-lift event aligned with the time window in which the drone lost altitude and crashed. The log indicated a combination of high blade AoA, negative vertical speed, and decreased throttle (<0.45) initially and then sharp rise (done by autopilot to stabilize the flight), with battery voltage trending downward. Disc loading was high due to payload, and the thrust-to-weight ratio approached its lower threshold, suggesting aerodynamic stress.

Figure 11 shows the throttle values over time, highlighting a sharp fall into the critical range just prior to the onset of stall. Figure 12presents the vertical speed, which transitions from a stable climb to a rapid descent, despite the autopilot continuing to command upward movement. Figure 13 displays the pitch angle, which correlates with changes in blade angle of attack and ultimately contributes to the stall event.

These trends confirm the vehicle was entering vortex ring state (VRS)—descending into its own downwash while maintaining high throttle—leading to turbulent airflow, loss of lift, and uncontrolled descent. This case highlights the importance of monitoring the interaction of flight parameters, not just isolated values—precisely the advantage of the proposed hybrid framework. It affirms the system's practical applicability and relevance in enhancing UAV operational safety.

E. IMPLICATIONS OF THE STUDY

The proposed stall prediction system has strong practical relevance across a wide range of aerial robotics applications—especially in scenarios involving stealth drones, military reconnaissance, and low-cost autonomous systems powered by companion computers like Raspberry Pi or NVIDIA Jetson.

In these missions, drones often operate in complex and uncertain environments where communication delays, low observability, and lack of GPS or human control limit real-time corrective action. Under such constraints, onboard systems must independently detect and react to critical flight instabilities like aerodynamic stall. This study provides a lightweight yet reliable stall detection framework that can be deployed directly on edge devices, without requiring cloud access or high-end GPUs.

By combining an efficient XGBoost-based prediction model with physically grounded rule-based logic, the system

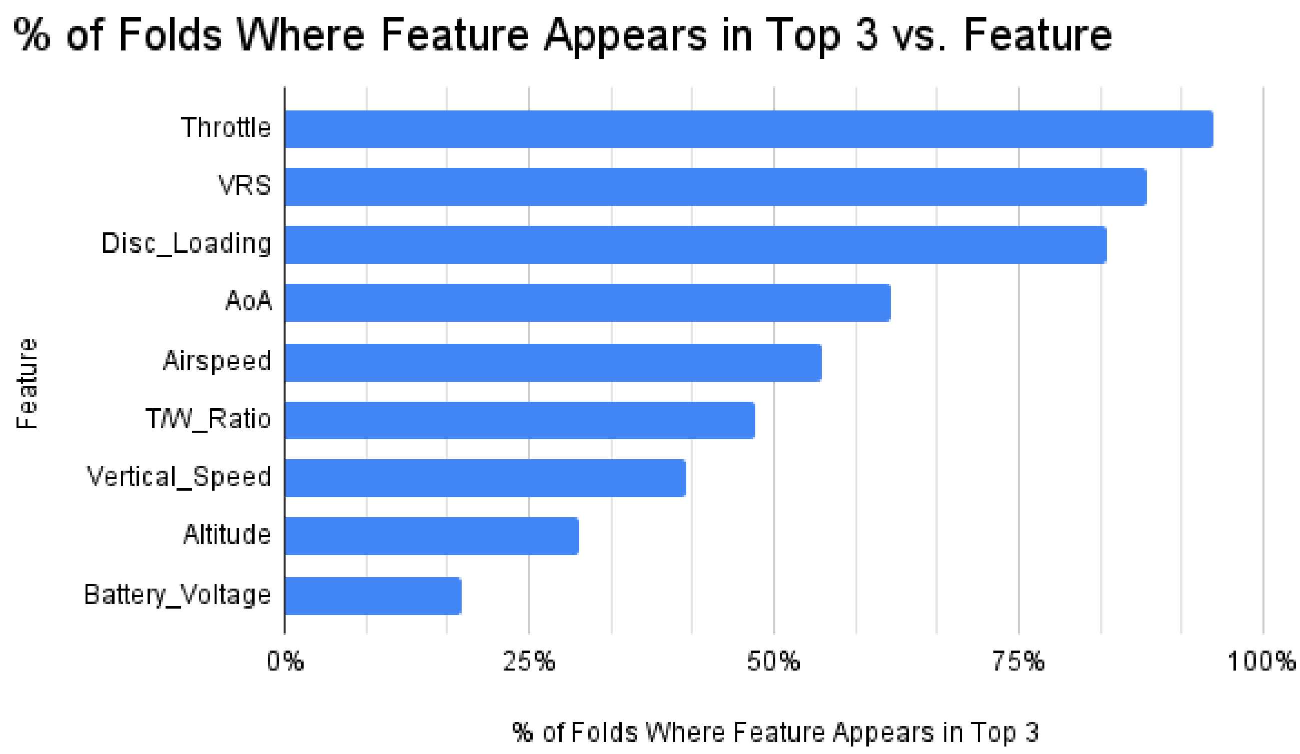

FIGURE 10. Feature Importance stability across cross-validation folds. The bar heights indicate the percentage of folds in which each feature appears among the top 3 most important according to SHAP values. Throttle, VRS, and Disc_Loading are consistently the top drivers of stall prediction, demonstrating the robustness of the model's attributions.

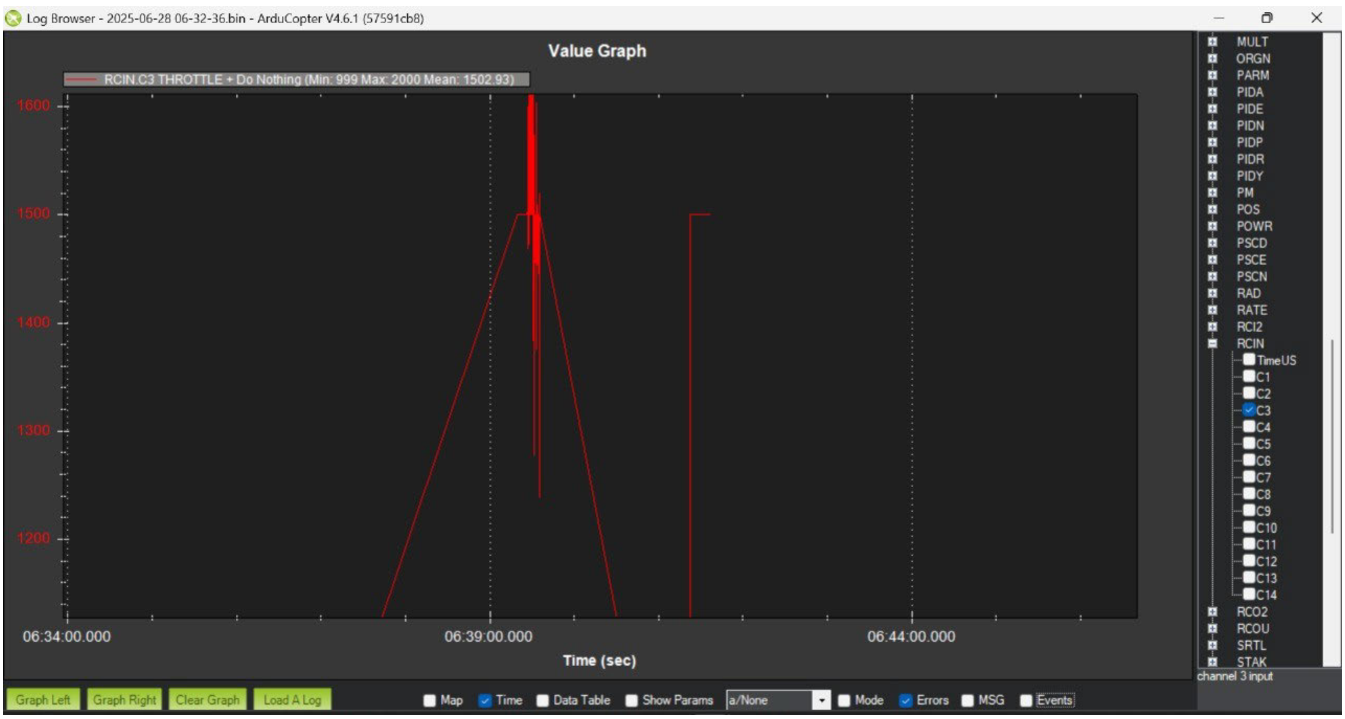

FIGURE 11. Throttle values over time showing a critical increase prior to stall onset.

FIGURE 12. Vertical speed profile revealing a transition from stable climb to sudden descent.

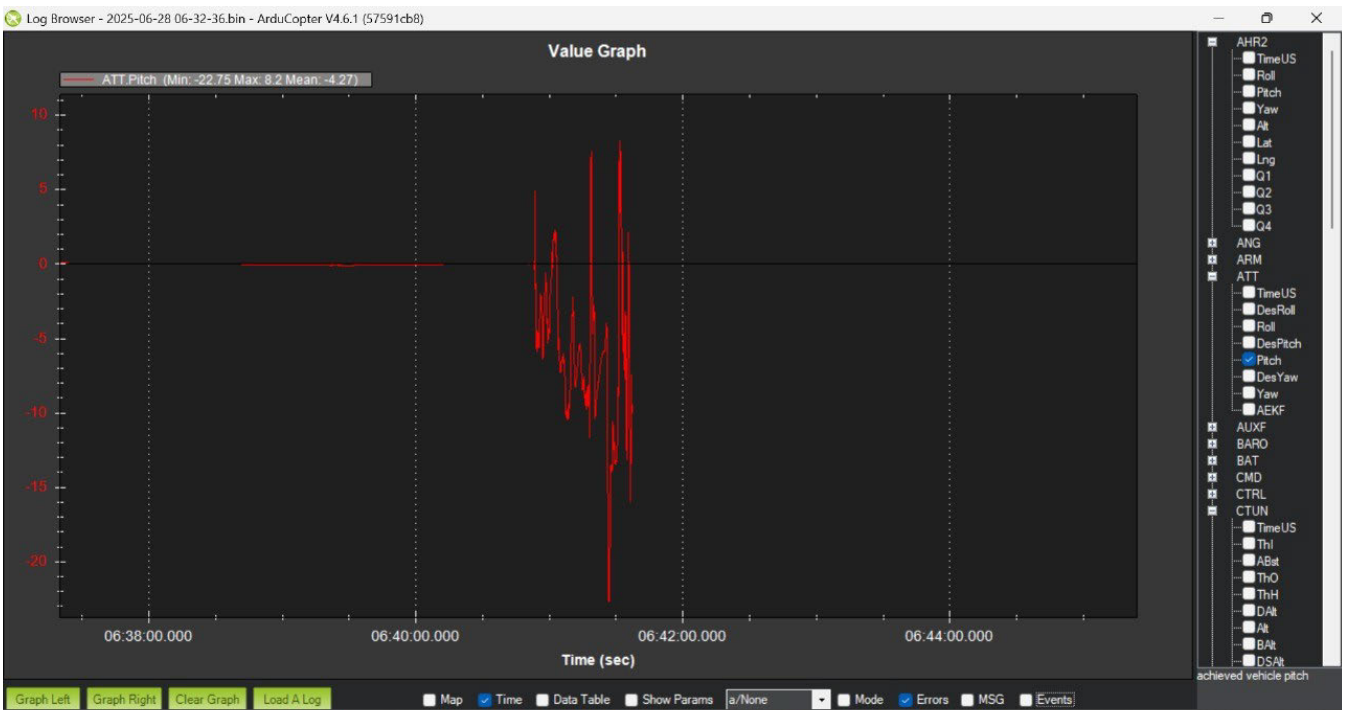

FIGURE 13. Pitch angle over time, contributing to changes in blade angle of attack.

ensures that even in cases where machine learning may underperform (e.g., in edge cases or noisy conditions), backup logic derived from flight physics continues to monitor for stall onset. This redundancy is crucial in military or stealth applications, where failure can lead to complete mission loss or vehicle crash behind enemy lines.

The explainability provided by SHAP values and clear aerodynamic thresholds (e.g., critical angle of attack, disc loading limits) ensures that operators and mission planners can understand why a stall alert was triggered. This is particularly useful during debriefing, mission validation, or regulatory audits—especially when autonomous drones are used in denied environments.

For swarm or fleet-based operations, this stall detection pipeline can be integrated into each drone's autonomy stack, enabling self-recovery behaviors such as load shedding, altitude stabilization, or emergency landing before full stall occurs. This improves the resilience of uncrewed aerial systems (UAS) in contested environments, such as border surveillance, urban mapping, or search and rescue missions in disaster zones.

Finally, the modular and retrainable nature of the proposed system allows rapid integration across a wide range of drone platforms—from lightweight quadcopters used for tactical ISR (Intelligence, Surveillance, Reconnaissance) to larger UAVs carrying sensitive payloads. Its low computational overhead and compatibility with onboard processors like Raspberry Pi make it well-suited for frugal innovation, where safety and autonomy must be achieved under hardware constraints.

The inclusion of SHAP-based explainability further enhances operational transparency by showing how each flight parameter contributes to stall predictions. This is especially critical in high-stakes or regulated environments where mission logs must justify autonomous decisions. The system's hybrid design—combining interpretable ML with rule-based safeguards—ensures reliability across edge cases and reduces both false positives and false negatives, as confirmed by ablation studies.

Overall, this research delivers a practical, deployable stall detection solution that supports real-time decision-making on embedded hardware, advancing drone autonomy for GPS-denied, communication-constrained, and mission-critical scenarios.

VI. CONCLUSION, LIMITATIONS, AND FUTURE SCOPE

This study introduces a hybrid stall prediction framework for quadcopters that integrates a machine learning model (XGBoost) with domain-specific rules to enable real-time, interpretable detection of aerodynamic instability. By simulating a comprehensive set of flight scenarios—including vortex ring state (VRS), rotor stall, and compound stress conditions—a structured dataset was generated to model stall behavior without risking hardware. The proposed system achieves both predictive performance and transparency, using SHAP-based explainability to highlight feature-level contributions and physics-based logic to enforce safety-critical checks (e.g., Blade AoA thresholds). Together, this allows for actionable and traceable stall alerts even during autonomous or communication-denied missions.

While the dataset used was synthetically generated, this was a deliberate choice to ensure safety, coverage, and control over variable space. Real-world data—including transient wind effects, actuator delays, or sensor drift—can be incorporated in future versions to improve ecological validity. Additionally, the current system favors deterministic rule integration over adaptive learning, but the modular architecture supports future extensions such as online retraining, federated learning, or hardware-in-the-loop validation.

The framework is optimized for deployment on companion computers like the Raspberry Pi, making it viable for resource-constrained missions such as tactical ISR, swarm-based surveillance, or stealth reconnaissance—where full GNSS or base-station feedback may not be available. It can be adapted to both frugal innovation platforms and high-end UAVs carrying sensitive payloads. In military, disaster relief, and contested airspace operations, this system provides early warning and interpretability, enabling safer flight autonomy and reducing mission aborts due to preventable aerodynamic failure.

Going forward, integrating environmental sensors, adaptive control response, and mission-specific tuning can elevate the framework into a plug-and-play stall prevention module—one that's portable across drone types and applicable in both civilian and defense sectors.

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